

Skills Summary

Out of What? And Who Got Asked?

Use this reference while completing your worksheet or when studying.

Skill 1 — Choosing the right denominator

Rule: $P(\text{event}) = \frac{\text{number of outcomes you want}}{\text{number of outcomes altogether}}$

- The denominator counts *everything that could happen*, including the outcomes you want. Never the leftovers on their own.
- If something is removed and not replaced, the total changes too — so re-check both numbers, not just the one the story mentions.
- Check: all the separate probabilities for one draw must add to 1.

Example: A drawer holds 7 red socks and 13 blue socks. One is taken without looking. Find $P(\text{red})$.

Step 1. Settle “out of what?” first: one sock is taken from the whole drawer, so the denominator counts every sock, red ones included.

Step 2. Altogether there are $7 + 13 = 20$ socks, and 7 of them are red, so $P(\text{red}) = \frac{7}{20}$.

Try it: A jar holds 9 mints and 6 toffees. One sweet is taken without looking. Find $P(\text{toffee})$ in lowest terms.

$\frac{9}{2}$:æp

(!) Watch out: $\frac{7}{13}$ compares red socks with blue socks. That is a *ratio*, not a probability, and it is always too big.

Skill 2 — Deciding whether a sample is fair

The test: could *every* member of the group you want to describe have been picked?

- If where or how you asked rules some people out, the sample is **biased** — and a bigger biased sample is still biased.
- Say which way it is wrong: asking at the bike rack over-states cycling.
- Scaling up a *fair* sample: $\text{estimate} = \text{population} \times \frac{\text{count in sample}}{\text{size of sample}}$.

Example: 25 students are drawn at random from a school roll of 400, and 10 of them walk to school. Estimate how many of the 400 walk.

Step 1. Names were drawn at random from the roll, so every student could have been picked — the sample passes the fairness test, and only then is it worth scaling up.

Step 2. The sample fraction is $\frac{10}{25}$, so the estimate is $400 \times \frac{10}{25} = \mathbf{160}$ students.

Try it: 40 students are drawn at random from a roll of 600, and 12 of them play an instrument. Estimate how many of the 600 do.

check: 081

Skill 3 — Finding a median from a sample

Rule: sort the values smallest to largest *first*, then take the middle one.

- An odd number of values has one middle value.
- An even number of values has two: the median is halfway between them.
- The median describes only the group you sampled. A median from a biased sample is a correct number about the wrong group.

Example: Find the median of 9, 4, 12, 7, 5, 10.

Step 1. A median is a position in the sorted order, so nothing can be read off the list as written — sort it before looking for the middle.

Step 2. Sorted: 4, 5, 7, 9, 10, 12. There are six values, so take the mean of the third and fourth: $\frac{7+9}{2} = 8$.

Try it: Find the median of 6, 2, 9, 3, 8, 4, 7.

check: 9